

various factors such as location, search patterns, etc. This gets users stuck in a bubble of repetitive content that they cannot get out of!

Unbubble solves this problem by gathering search results from as many sources as possible so that users may receive truly versatile and neutral search results.

Unbubble is committed to protecting its users' privacy. It also hides the origin of visitors by removing the referrer field from HTTP headers.

Neutrality and uncensored results are the core principles of Unbubble, so each search result drawn from the search providers is assessed based on neutrality and relevance, and the results that are more detailed and neutral are placed at the top.

Similar to DuckDuckGo's Bangs, Bubble tags are like shortcuts to particular websites. Some popular websites have more than one Bubble tag. Unbubble uses its own auto-completion system with strong SSL; so search terms are protected from outside parties.

Search operators are used for retrieving more relevant results by supplying additional instructions to the search queries; for example, '+cat' retrieves the results containing exactly the keyword 'cat' and the opposite happens when '-cat' is used, i.e., the search results will not contain the 'cat' keyword in them.

Search Encrypt

Search Encrypt is a search engine that uses a unique approach when it comes to privacy. Like the other search engines featured in this article, it does not track its users or their search terms. Unlike other search engines, however, it uses encryption on search queries made by the user using a short-lived key before the queries are sent to the Search Encrypt servers. So if you look at your search history, you

will see that your search queries are encrypted. After 30 minutes of inactivity, your Search Encrypt search result pages expire — so no one will ever know what search terms you entered.

It also uses the Perfect Forward Secrecy method to ensure that a new key is generated for each session. If a session key that is used in data transmission is compromised, it can reveal only the data pertaining to that key alone. So, basically, if an attacker manages to get to know a session key, it cannot be used to decrypt your previous sessions.

Some search engines mentioned here display ads in their search results since they have to generate revenue, but none of them use these ads to violate the user's privacy. Each search engine is unique so it is up to the users to select the right one for their needs. **END** 

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The Bleachbit utility

One very useful multi-purpose tool for cleaning the system and increasing privacy is Bleachbit. It is an open source cross-platform application available on various distributions of Linux and even Windows. It is a good tool for those who are used to CCleaner in Windows. It had its moment of fame when it came to be known that Hillary Clinton's IT team used Bleachbit to wipe the email servers clean so that no data could be recovered.

Bleachbit can be used to clean up the junk files and folders of commonly used applications, delete temporary files and free up space. It also clears the browser cache, the list of downloaded files and even the commands in the *History* file. In addition, it can be used to shred files and folders so that they cannot be recovered and wipe the free disk space.

Bleachbit can be installed by using the following command in an Ubuntu system:

```
$sudo apt-get install bleachbit
```

The utility can be run both in CLI and GUI mode.

Once the program opens in GUI mode, the left side displays the list of folders and files to clean up, which can then be checked by the user. Click on the 'x' icon above the list to remove the files. You can click on the magnifying glass icon to see the list of files that will be removed. It is better to avoid the package section, and do not run it as the root user unless you are an experienced user.

Shredding using Bleachbit: To shred files or folders, click on the *File* menu on top and choose either *Shred files* or *Shred folders* after which you can choose the items to be shredded. The space occupied by the files/folders is then completely overwritten so that the original contents cannot be recovered. **END** 

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